

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Study Title: Study of Boserolimab (MK-5890) as Monotherapy and in Combination With Pembrolizumab (MK-3475) in Adults With Advanced Solid Tumors (MK-5890-001) **Official Title:** A Phase 1 Study of MK-5890 as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced Solid Tumors **Short Title/Acronym:** MK-5890-001 **Protocol Number:** MK-5890-001 (IND: 136304) **NCT Number:** NCT03396445 **Sponsor Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Boserolimab (MK-5890) [anti-CD27 agonist], Pembrolizumab (MK-3475 / Keytruda), Pemetrexed, Carboplatin, Nab-paclitaxel **Phase of Development:** PHASE1 **Trial Status:** COMPLETED **Medical Monitor:** Clinical Director, Merck Sharp & Dohme LLC

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of boserolimab (MK-5890) as monotherapy and in combination with pembrolizumab (and chemotherapy), and to determine the Dose-Limiting Toxicities (DLTs) and adverse events (AEs). **Secondary Objectives:** To evaluate preliminary antitumor activity and characterize the pharmacokinetics of boserolimab (including AUC_{0-last}, AUC_{0-3w}, and AUC_{0-6w}).

2.2 Background & Rationale To address the unmet medical need in patients with advanced solid tumors, including Endometrial Cancer, Carcinoma, Triple-Negative Breast Neoplasms (TNBC), and Non-Small-Cell Lung (NSCLC) cancer. Boserolimab is an investigational humanized anti-CD27 agonist designed to stimulate T-cell immunity. This study evaluates whether adding CD27 co-stimulation to PD-1 blockade (pembrolizumab) and chemotherapy improves immune responses and clinical outcomes in patients with advanced malignancies. The study explores both safety and pharmacokinetics.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose Escalation / Cohort Expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 182 evaluable patients across all dose escalation and expansion cohorts. **Number of Sites:** Multicenter (Global) **Estimated Study Start:** March 2018 **Study Status:** COMPLETED

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

1. Age ≥ 18 years with an Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1.
2. **Arms 1 & 2:** Histologically or cytologically confirmed advanced/metastatic solid tumor by pathology report and has received or been intolerant to all treatment known to confer clinical benefit.
3. **Arm 3:** Histologically or cytologically confirmed diagnosis of stage IV (M1a or M1b per current AJCC criteria) non-squamous NSCLC.
4. **Arm 4:** Triple-negative breast cancer (TNBC) that is locally recurrent, inoperable, not previously treated with chemotherapy, and which cannot be treated with curative intent OR metastatic disease not previously treated with chemotherapy.
5. Measurable disease by RECIST v1.1. Target lesions situated in a previously irradiated area are considered measurable if progression has been demonstrated in such lesions.
6. Adequate organ function.
7. Male participants must agree to use adequate contraception and refrain from donating sperm during the treatment period and for at least 120 days after the last dose of boserolimab/pembrolizumab OR 180 days after chemotherapeutic agents. Female participants must not be pregnant or breastfeeding and must use adequate contraception for the same timeframes.
8. Submit an evaluable baseline tumor sample for analysis (newly obtained or archival).

4.2 Exclusion Criteria

1. History of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years.
2. Clinically active central nervous system (CNS) metastases and/or carcinomatous meningitis.
3. Severe hypersensitivity reaction to treatment with a monoclonal antibody (mAb) and/or other components of the study treatment.
4. Active infection requiring systemic treatment, known HIV, and/or active and acute Hepatitis B or C infections.
5. History of interstitial lung disease or (noninfectious) pneumonitis that required steroids, or current pneumonitis.
6. Symptomatic ascites or pleural effusion.
7. Previously had a stem cell, bone marrow, or solid organ transplant.
8. Active autoimmune disease requiring systemic treatment in the past 2 years (except vitiligo or resolved childhood asthma/atopy).
9. Not fully recovered from any effects of major surgery without significant detectable infection.
10. Pregnant, breastfeeding, or expecting to conceive/father children.
11. Had chemotherapy, definitive radiation, or biological cancer therapy within 4 weeks (2 weeks for palliative radiation) before first dose, or has not recovered to Grade ≤ 1 from AEs.
12. Expected to require other antineoplastic therapy.
13. On chronic systemic steroid therapy in excess of replacement doses (>10 mg/day prednisone equivalent) or immunosuppressive medication.
14. Regular illicit drug use or recent history (within last year) of substance abuse.
15. Received a live-virus vaccine or participated in a study of an investigational agent/device within 28 days before the first dose.
16. **Additional for Arm 3:** Received lung radiation >30 Gy

within 6 months; unable to interrupt aspirin/NSAIDs for 5-day (8-day for long-acting) period; unable/unwilling to take folic acid or vitamin B12 supplementation. 17. **Additional for Arm 4:** Hypersensitivity to nab-paclitaxel; neuropathy $\geq$ Grade 2; history of class II-IV CHF or MI within 6 months; previous treatment with immune checkpoint inhibitor(s) (e.g., PD-1/PD-L1).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Monotherapy:** Boserolimab IV administered at escalating doses. Participants experiencing disease progression may switch to combination therapy for up to 35 cycles at investigator discretion. * **NSCLC Cohort (Arm 3):** Boserolimab + Pembrolizumab + Pemetrexed + Carboplatin. * **TNBC Cohort (Arm 4):** Boserolimab + Pembrolizumab + Nab-paclitaxel. **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Up to 35 administrations/cycles (approximately 2 years), or until disease progression, unacceptable toxicity, or investigator decision.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care as clinically indicated. Participants in Arm 3 must take folic acid/B12. **Prohibited:** Live-virus vaccines within 28 days of first dose, systemic corticosteroids for immunosuppression (>10 mg/day prednisone equivalent), concurrent investigational agents, and illicit drugs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging (RECIST baseline), Labs, Tumor Sample
Baseline	Day 1	First Dose of MK-5890/Pembrolizumab/Chemo, PK Sampling
Interim	Q3W or Q6W	Safety Labs, irAE Assessment, PK Draws, Tumor Imaging every 9 weeks
End of Study	120-180 Days Post-Tx	Final Vital Signs, AE Resolution tracking, Contraception verification, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- Number of Participants Who Experienced a Dose-Limiting Toxicity (DLT):** Assessed via CTCAE v5.0 (e.g., Grade 4 nonhematologic toxicity, Grade 4 hematologic toxicity lasting ≥ 7 days, Grade 3/4 nonhematologic lab abnormality, missing $>25\%$ of study drug during evaluation, etc.).
- Number of Participants With One or More Adverse Events (AEs):** Any untoward medical occurrence temporally associated with the use of study treatment.
- Number of Participants Who Discontinued Study Treatment Due to an AE.**

7.2 Secondary Outcomes 1. Area Under the Concentration-Time Curve From Time 0 to Last Quantifiable Concentration (\$AUC_{0-last}\$) of Boserolimab. 2. Area Under the Concentration-Time Curve From Time 0 to 3 Weeks (\$AUC_{0-3w}\$) of Boserolimab. 3. Area Under the Concentration-Time Curve From Time 0 to 6 Weeks (\$AUC_{0-6w}\$) of Boserolimab.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite clinical visits, physical exams, and lab assessments, evaluated via CTCAE v5.0. **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal Safety Review Committee continuously evaluates dose-escalation thresholds, DLTs, and expansion cohort safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic outcome measures will not be analyzed separately for switch-over treatment arms. **Sample Size Justification:** Targeted enrollment of 182 participants to establish safety profile, MTD/RP2D, and provide precision around PK and safety estimates. **Handling of Missing Data:** Missing efficacy evaluations are generally treated as non-responders. Safety data is evaluated using descriptive statistics without imputation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote source data verification conducted by the sponsor's clinical monitoring team.

Audit Trail: Fully validated Electronic Data Capture (EDC) system with secure audit trails and data clarification form (DCF) procedures.

11. Study Identifiers & Governance

Primary ID: MK-5890-001 **Registry Number:** NCT03396445 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** Study of Boserolimab (MK-5890) as Monotherapy and in Combination With Pembrolizumab (MK-3475) in Adults With Advanced Solid Tumors (MK-5890-001) **Official Regulatory Title:** A Phase 1 Study of MK-5890 as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced Solid Tumors

12. Clinical Framework (Oncology Specific)

Disease Areas / Conditions: Pharmacokinetics, Solid Tumor, Carcinoma, Non-Small-Cell Lung (NSCLC), Triple Negative Breast Neoplasms (TNBC), Endometrial Cancer. **Diagnosis Threshold:** Histological or cytological confirmation of advanced, locally recurrent, inoperable, or metastatic disease. **Medical Methodology:** Immuno-oncology Combination Therapy (anti-CD27 + anti-PD-1 + Chemotherapy including Nab-paclitaxel, Pemetrexed, and Carboplatin). **Phase:** PHASE1 (Completed)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway augmented with investigational therapy algorithms. **Education Protocol (HCP):** Site Initiation Visits (SIV) covering protocol procedures, dose modification, and toxicity management guidelines. **Education Protocol (Patient):** Informed consent process, contraception guidelines (up to 180 days post-chemo), and guidance on identifying toxicities. **Quality Audit Mechanism:** Routine clinical site monitoring and central statistical data checks.

14. Observation & Follow-Up Schedule

Total Duration: Continued treatment for up to 35 administrations/cycles (~2 years) followed by long-term survival and contraception follow-up. **Onsite Visit Cadence:** Depending on cohort: typically Q3W or Q6W cycles. **Remote Monitoring Frequency:** Adverse event and survival tracking every 12 weeks post-treatment. **Checklist Review Interval:** Tumor imaging assessments (CT/MRI) approximately every 9 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				